Lab 1 Outline Final Draft

Team Diamond

1. Introduction

This section was written by: Priyaja Surendra

Societal Problem

1. Children spend a significant amount of time using digital devices for entertainment.
2. In 2021, children between the ages of 8 and 12 averaged approximately five hours and 33 minutes of entertainment screen use per day, compared with four hours and 44 minutes in 2019 (Rideout, 2022).
3. Excessive or unproductive technology use may negatively affect children's cognitive development, problem-solving skills, social interactions, and mental and emotional health (Lynn University, 2024).
4. Parents need an effective way to establish healthy screen-time limits while encouraging their children to use technology productively.

Needed Solution Characteristics

1. The solution should allow parents to establish and manage screen-time limits.
2. It should encourage children to complete educational activities.
3. Children should be able to earn additional screen time by completing educational tasks.
4. Parents should be able to customize questions and rewards according to their child's needs.
5. The solution should track each child's educational activity and progress.

Proposed Product

1. EduQuest is an educational screen-time management application designed for parents and children.
2. It allows children to earn additional screen time by completing educational questions and activities.
3. Parents can use EduQuest to manage screen-time limits, select or create educational questions, establish rewards, and review their child's progress.

4. The proposed prototype will demonstrate EduQuest's primary features, including the Parent Controls Dashboard, educational questions, screen-time rewards, and progress tracking.

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2. EduQuest Product Description

This section was written by: Tennya Boone

EduQuest is a parent-controlled screen-time management and learning platform designed for school aged children and their parents. This solution helps transform recreational screen time into an opportunity for learning by requiring children to complete short educational questions or activities before receiving any additional screen time on their device. Parents would be able to set screen-time limits, choose the subjects and the questions their child receives, and customize how screen time can be earned. EduQuest would also provide explanations when children answer questions incorrectly and allow parents to monitor their child's educational activity and progress through a parent dashboard for easy use. These features work together to make screen time more productive for children while giving parents more control over how their children use their devices in day-to-day life. For this prototype, screen-time limits are enforced within the EduQuest app itself rather than at the operating system level. This keeps the initial build achievable within the scope of a class project while still demonstrating the core parent-child, question-and-reward mechanic.

The main goal of EduQuest is to encourage children to spend some of their recreational screen time strengthening the necessary academic and problem-solving skills. This platform is designed to address the problem of children spending large amounts of time using devices for entertainment while important, foundational academic performance levels continue to decline. By ensuring that learning comes before entertainment with these required educational activities, EduQuest aims to turn an activity children already enjoy into an opportunity for learning instead of simply restricting their screen time altogether. The platform also aims to help parents manage their children's screen use and understand their educational progress without completely preventing children from having access to their devices for entertainment.

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2.1. Key Product Features and Capabilities

This section was written by: Sincere Sanders

Significance

EduQuest transforms screen time from something that is simply restricted into a read based educational engagement tool. According to an article from Lynn University, one potential disadvantage of technology at a young age is that it “potentially diminishes cognitive development and reduces problem solving skills” (Lynn University, 2024). EduQuest address this concern by requiring children to complete educational questions before earning additional screen time. This encourages children to spend more time learning while giving parents control over their child’s screen time limits. This application is unique because it combines education, parental controls, and screen time rewards all into one system, making screen time an incentive for productive behavior rather than a source of concern.

Main Features & Capabilities

- **Parental Controls** - The Parent Controls Dashboard will allow the parent to set screen time limits for their child, determine how screen time can be earned, as well as the amount of extra time given for the completion of educational tasks. This allows parents to customize EduQuest to their child's needs.
- **Educational Questions** - EduQuest will provide a built in database of educational questions covering a variety of subjects. Parents can also choose to use the questions provided by EduQuest or create their own questions using the Parent Control Dashboard.
- **Screen Time Rewards** - This rewards system gives children additional screen time after they complete the required educational activities. This feature creates a direct connection between completing educational work and receiving a reward.
- **Progress Tracking** - This progress tracking feature will record the child’s educational activity and performance, allowing the parent to monitor their child’s progress over time. This gives parents insight into how their child is

performing and allows them to adjust settings in their dashboard based on their child's performance and needs.

2.2. Major Components (Hardware/Software)

This section was written by: Luke Morken

Note: *An earlier version of this design included OS-level enforcement (iOS Screen Time API, Android Device Admin API) to lock the device itself. We have removed this from the current prototype, since both platforms require developer permissions and formal app review/approval from Apple and Google to access these APIs — a process that is not feasible within our project timeline. We view OS-level enforcement as a natural direction for future development if the product moves toward a commercial or extended release, since it would allow EduQuest to restrict access to other apps and the device as a whole, not just its own interface.*

HARDWARE

1. **Client Devices (End-user Hardware)**

    - iOS and Android smartphones/tablets running the React Native or Flutter app
    - Requires internet connectivity (wi-fi/cellular), push notifications (APNs for iOS and FCM for Android)

2. **Backend Server Infrastructure (Fast API/Django + Docker Block)**

    - Application servers or containers hosting the Dockerized Python API
    - Minimum viable: 1 server instance (2 cCPU / 4gb RAM) for development and small scale deployment
    - For production, horizontally scaled instances with a load balancer for when user count grows
    - Can be run on cloud or on-premises (AWS EC2, Google Cloud Run, Azure App Service, etc.). Cloud is the preferred method for a our student project as a way to simplify deployment

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3. **Database Server**

    - Dedicated or managed database instance (AWS RDS or a Postgres container on the same host for MVP). We will do Postgres container because it is simpler for our prototype
    - Storage sized for user profiles, activity/usage logs, and relational data between the parent and child accounts
    - Will need persistent disk storage with backup and or snapshot capabilities

4. **Messaging Infrastructure (Needs to Support Firebase Messaging Block)**

    - We won't need dedicated hardware because we are using Google Firebase Cloud messaging, which is run and maintained by Google
    - Server-side integration point only; Firebase handles the delivery infrastructure

5. **Network Layer**

    - HTTPS/TLS termination (load balancer or reverse proxy like Nginx) in front of our API server
    - Standard internet-facing infrastructure. We don't need specialized networking hardware since there isn't low level device management traffic we'll need to secure

SOFTWARE

1. **Mobile client app using React Native or Flutter.**

    - This app will be a cross-platform codebase(iOS and Android) that will serve as both the app the child uses and the parent's dashboard.

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2. **Backend API server using Fast API or Django.**

    - It will handle authentication, business logic, parent-child linking, and will serve data to the mobile client over REST/HTTPS

3. **Database Layer using PostgreSQL.**

    - It will store the user account and parent-child relationships and data.

4. **Notification service integration using Firebase Cloud Messaging.**

    - It will send push notifications to parents (usage summaries, number of questions asked, questions answered correctly vs incorrectly, etc.) We have removed OS-level enforcement so we won't need to alert the parents about a child attempting to delete the app. It will send push notifications to parents (usage summaries, number of questions asked, questions answered correctly vs. incorrectly, etc.). Because this prototype does not include OS-level enforcement, EduQuest cannot detect or alert parents to attempts to delete or bypass the app at the system level. This capability would require the OS-level permissions described above and is a potential future enhancement.

5. **Deployment using Docker + Linux.**

    - Docker will put our API into a container so that it can consistently deploy across different environments (iOS and Android and or future environments).

Note: *While EduQuest's current prototype cannot manage a child's screentime or access, a future version could incorporate OS-level enforcement using Apple's Screen Time API (iOS) and Android's Device Admin API. These APIs would allow EduQuest to restrict access to other applications and the device as a whole rather than just its own interface, more closely matching tools like Apple Screen Time or Qustodio referenced in Section 3. Implementing this would require developer entitlements and formal app review from Apple and Google, which fall outside the scope of this prototype.*

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3. Identification of Case Study

This section was written by: Justin Carmona

Primary Users

EduQuest is being developed for parents of children ages 8 to 12 who want a way to balance recreational screen time with academic reinforcement, without resorting to blunt tools like hard time limits or full app blocking. These are parents who already care about limiting screen time using tools like Apple Screen Time, Qustodio, or Bark, but are frustrated that existing options only restrict and not give a productive alternative or build in any academic upside. The secondary user is the child themselves, who interact directly with the question-answering interface to unlock additional screen time.

Why this group?

This age range is being targeted because it aligns with the elementary/upper-elementry years where foundational reading and math skills are still forming, and where screen time is heaviest and least supervised (Rideout, 2022). Parents in this group are actively making screen–time decisions daily, making them a motivated buyer, but current tools on the market do not combine content filtering/time management with an academic earning mechanic. That is the gap that EduQuest fills.

Future / Expanded Users

Teachers or Tutors - Down the line, teachers could build or assign custom question sets tied to what's actually being covered in class that week, turning EduQuest into a home-reinforcement or test-prep tool rather than generic practice. A teacher-facing dashboard could show which students are consistently missing certain question types, flagging where extra help is needed.

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Homeschool Families - Homeschool parents already act as both parent and educator, so they'd benefit from EduQuest's dual role: managing screen time and delivering self-paced,gradually adapting practice content. Progress tracking could double as informal record-keeping for their curriculum.

Expanded Age Ranges - The core mechanic of EduQuest is not inherently tied to ages 8-12. Content could scale down to early readers (K-2, sight words, basic counting) or up to middle schoolers (pre-algebra, reading comprehension), letting EduQuest grow with a family rather than being outgrown after a few years.

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4. Glossary

This section was written by: Jessica Melton

Term	Definition
Academic Reinforcement	A practice that helps strengthen educational concepts a child is already learning.
API (Application Programming Interface)	A way for different parts of a software system to communicate and exchange information ("API").
Backend Server	The part of the system that handles data, requests, and other processes behind the scenes.
Built-in Question Database	A collection of educational questions provided by EduQuest.
Cloud Computing	Using internet-based servers and services to run applications and store data (Mell and Grance, 2011).

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Container	An environment that contains an application and what it needs to run effectively (“What Is a Containter”).
Cross-Platform	Software that is designed to work on more than one operating system (“Build for and Integrate”).
Custom Questions	Custom educational questions created by parents.
Docker	A platform used to package and run software in containers so it can work across different environments (“What Is a Docker”).
FCM (Firebase Cloud Messaging)	A Google service that is used to send notifications to devices and applications (Firebase Cloud Messaging).
Flutter	A service used to build applications that can run on multiple platforms using one codecase (“Build for and Integrate”).
Load Balancer	A tool that sorts incoming requests across multiple servers to prevent one server from handling too much traffic (“What is Elastic Load Balancing”).
Parent Control Dashboard	The main area where parents manage screen-time limits, questions, rewards, and their child's progress.
Parental Controls	Settings that allow parents to manage how their child uses EduQuest, including screen-time limits and rewards.
PostgreSQL	The database system EduQuest will use to

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	store account information and other application data (“About PostgeSQL”).
Progress Tracking	A feature that records a child’s educational activity and performance over time.
React Native	A service used to build mobile applications for platforms using one codebase (“React Native”).
Recreational Screen Time	Time spent using digital devices mainly for entertainment, such as games or videos.
REST	A common way for applications to request, send, and receive information through an API (“REST”).
Screen-Time Limit	The amount of recreational device time a parent allows a child before additional time must be earned.
Screen-Time Management	The process of setting and managing how much recreational time a child can spend on a device.
Screen-Time Rewards	Additional device time a child earns by successfully completing required educational questions and activities.
TLS (Transport Layer Security)	Technology used to protect information when it travels over the internet (Rescorla).

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